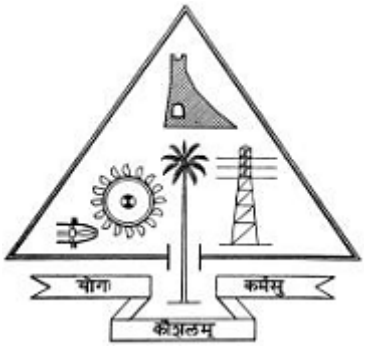
ONLINE FILING OF RTI SECOND APPEAL AND PROCESSING AT STATE INFORMATION COMMISSION OF KERALA(RASIC)

Thesis submitted in partial fulfillment of the requirements for the award of the degree of ***Master of Computer Applications*** *of the* ***University of Calicut***

submitted by

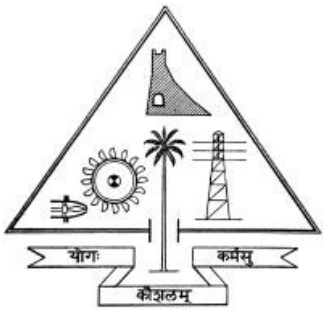
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DEPARTMENT OF COMPUTER APPLICATIONS
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NOVEMBER 2018

DEPARTMENT OF COMPUTER APPLICATIONS
GOVERNMENT ENGINEERING COLLEGE, THRISSUR
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CERTIFICATE

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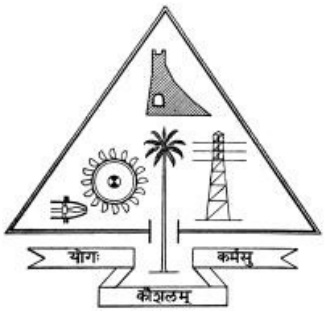
under my supervision and guidance, and is submitted in November 2018 in partial fulfillment of the requirements for the award of the Degree of Master of Computer Applications from APJ Abdul Kalam Technological University(KTU).

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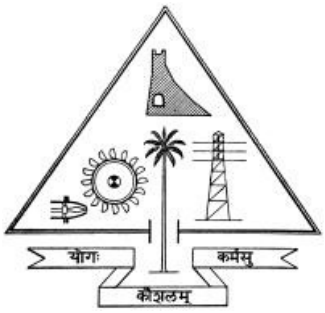
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DECLARATION

I hereby declare that the thesis titled, **Online Filing of RTI Second Appeal and Processing at State Information Commission Kerala(RASIC)**, is my own work and that, to the best of my knowledge and belief, it contains no material previously published or written by another person nor material which has been accepted for the award of any other degree or diploma of the university or any other institute of higher learning, except where due acknowledgement and reference has been made in the text.

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ABSTRACT

This project work is towards the design and development of workflow driven web portal covering the various life cycle activities of second appeal of RTI act.Any citizen who is not satisfied with the information provided is not provided against the RTI petition filed in any government department after the first appeal can file a second appeal with the state information commission(SIC).RASIC is web based application follow various workflows and cover functionalites of second appeal processing at NIC.The software is developing in NIC,which is develop and developing e-Gov IT solution to State Information Commission(SIC),Kerala

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NOMENCLATURE

RTI	Right To Informations
PIO	Public Information Officer
FAA	First Appellate Authority
SIC	State Information Commission
NIC	National Informatics Centre
ICT	Information and Communications Technology
NICNET	National Informatics Centre NETwork or NIC network
NKN	National Knowledge Network
IDE	Integrated Development Environment
JDT	Java Development Ttools
MVC	model view controller
ORM	Object Relational Mapping
OWASP	Web Application Security Project
CERT	Computer Emergency Response Team
KSWAN	Kerala State Wide Area Network
GIGW	Guidelines on Indian Government Website
SPD	Software Procedural Design
DBMS	Database Management System
BCNF	Boyce Codd normal form
DFD	Data Flow Diagram
MIS	Management Information System
LDAP	Lightweight Directory Access Protocol
CRLF	Carriage Return and Line Feed
SAST	Static application security testing
XSS	Cross-Site Scripting

CHAPTER 1
INTRODUCTION

1.1 Right To Information

The Right to Information Act [1] is considered to be one of the strongest pieces of legislation at the disposal of the common man. RTI gives citizens the power to question public authorities and their activities, thereby not only promoting transparency, but demanding accountability as well. The Act is considered a landmark legislation in the fight against corruption.

Right to Information (RTI) is an Act of the Parliament of India to provide for setting out the practical regime of the right to information for citizens and replaces the erstwhile Freedom of information Act, 2002.

Any citizen of India may request information from a ”public authority” (a body of Government or ”instrumentality of State”) which is required to reply expeditiously or within thirty days. The Act also requires every public authority to computerize their records for wide dissemination and to proactively certain categories of information so that the citizens need minimum recourse to request for information formally.

RTI covers all the constitutional authorities, including executive, legislature and judiciary; any institution or body established or constituted by an act of Parliament or a state legislature.

National Informatics Center has already introduced workflow based RTI system for basic services and first appeal as a part of state eDistrict Mission Mode Project,with the state IT mission.

1.1.1 RTI Application

The RTI Act 2005 mandates Public Information Officer (PIO) must provide the information as asked by the applicant within 30 days of receipt of RTI application.

Public information officers (PIOs) are the communications coordinators or spokespersons of governmental organizations The primary responsibility of a PIO is to provide information to the public and media as necessary and to meet the legal requirements.

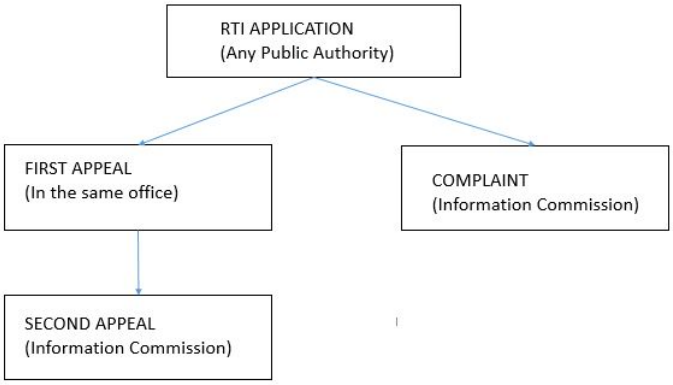


Fig. 1.1: RTI

1.1.2 RTI First Appeal

The information sought by an applicant should either by supplied to him or his application should be rejected within the time prescribed by the Act.[5] If additional fee need by charged from the applicant, communication in this regard should be sent to him within the time limit prescribed for sending information. If the applicant does not receive information or decision about rejection of request or communication about payment of additional fee within the specified time, he can make an appeal to the First Appellate Authority. Appeal can also be made if the applicant is aggrieved by the decision of the PIO regarding supply of information or the quantum of fee decided by the PIO.

First Appeal should be filed before the designated First Appellate Authority (FAA), who is an officer higher in rank than PIO.Details of First Appellate Authority shall be included in PIO's decision letter itself. When there is no response / decision from PIO, it is better to search website (RTI link) of the public authority for FAA details.

The first appeal can be filed when the following thing happens:

- Information is not received within 35 days (5 days extra for post-delivery etc.) of the date of filing of the RTI application.
- Information received is incomplete or you are not as asked.
- RTI was transferred to another department with intimation to you but a reply is not received even after 35-40 days.

1.1.3 RTI Second Appeal

Anybody who is unhappy or dissatisfied with the decision of the Appellate Authority can file Second Appeal to the Information Commission at the Centre or respective States.[2] For matters related to State Government public authorities, send your appeal to concerned State Information Commission.

Appeals against panchayats should also be sent to the relevant State Information Commission. The Information Commission may admit the 2nd appeal after the expiry of the period of 90 days The State Information Commission has the power to require the Public Authority to comply with the Provisions of the RTI Act

1.1.4 Scope

This project work is towards the design and development of workflow driven web portal covering the various life cycle activities of second appeal of RTI act.Any citizen who is not satisfied with the information provided is not provided against the RTI petition filed in any government department

after the first appeal can file a second appeal with the state information commission(SIC).

1.1.5 State Information Commission

Vision of SIC [5] is provide for setting out the practical regime of right to information for citizens to secure access to information under the control of public authorities, in order to promote transparency and accountability in the working of every public authority. The Powers and Functions of the Information Commission are enumerated under chapter 5 of the RTI Act .

1. Enquiry into Complaints

   It shall be the duty of the Commission to receive and inquire into complaints from persons aggrieved by any of the reasons given under section 18 (1) of the Act. Where the Commission is satisfied that there are reasonable grounds to inquire into the matter, it may initiate an inquiry thereof. While conducting an inquiry into a complaint, the Commission shall have the same powers as are vested in a civil court while trying a suit under the Code of Civil Procedure, 1908, in respect of the matters given under section 18 (3) of the Act. During an inquiry into a complaint under this Act, the Commission can examine any record to which the RTI Act applies which is under the control of the Public Authority.

2. Deciding 2nd Appeals

   Section 19(3) of the RTI Act provides for 2nd appeal before the Central/State Information Commission. The 2nd appeal shall lie within 90 days from the date on which the decision should have been made by or was actually received from the first appellate authority. The Information Commission may admit the 2nd appeal after the expiry of the period of 90 days, if it is satisfied that the appellant was prevented by sufficient cause from filing the appeal in time. While making a decision, the State Information Commission has the power to require the

Public Authority to comply with the Provisions of the RTI Act -

- by providing access to information
- by appointing the State Public Information Officer
- by publishing certain information or categories of information
- by making necessary changes to its practices in relation to the maintenance, management and destruction of records
- by enhancing the provision of training on the right to information for its officials

The second appeals filed before the State Information Commission are decided in accordance with the Kerala State Information Commission (Procedure for appeal) Rules, 2006, notified by the Government of Kerala as No.27774/Cdn.5/2006/GAD dated 31.5.06 . In an appeal proceedings, the onus to prove that the denial of request was justified shall be on the Public Information Officer who denied the request . If the appeal relates to information of a 3rd party, the 3rd party shall be given a reasonable opportunity of being heard before a decision is made by the Information Commission.

3. compensation

   The RTI Act empowers the Information Commission to require the Public Authority to compensate the complainant for any loss or other detriment suffered.

4. Imposing Penalties

   The State Information Commission has the power to impose penalty on the State Public Information Officer for the following defaults:

   (a) without reasonable cause refused to receive an application for information or has not furnished information within the specified time limit

   (b) malafidely denied the request for information

(c) knowingly given incorrect, incomplete or misleading information

(d) destroyed information which was the subject of the request

(e) obstructed in any manner in furnishing the information.

The Penalty provided under section 20(1) of the RTI Act is Rs.250/- each day till the application is received or information is furnished. The total amount of such penalty shall not exceed twenty five thousand rupees.

5. Disciplinary Action

Where the State Information Commission, at the time of deciding any complaint or appeal is of the opinion that the State Public Information Officer has, without any reasonable cause and persistently, failed to receive an application for information or has not furnished information within the time specified under sub-section (1) of section 7 or malafidely denied the request for information or knowingly given incorrect, incomplete or misleading information or destroyed information which was the subject of the request or obstructed in any manner in furnishing the information, the Commission can recommend for disciplinary action against the State Public Information Officer, under the service rules applicable to him under section 20(2) of the Act.

1.2 Objective

The objective of the project work is to design and develop workflow based a web application to handle second appeal of RTI application at SIC Kerala

This web application will allow the citizen to file RTI second appeals to Kerala state information commission(SIC)of Kerala online.

Online processing of received appeal at State Information Commission also will be addressed by the system.This web portal will be integrate with the RTI portal of the govt. of Kerala under the eDistrict platform.

CHAPTER 2
ENVIRONMENTAL STUDY

2.1 System Configuration

System configuration describe the hardware and software requirement of the system for development

2.1.1 Hardware Requirements

- Memory : 8 GB of RAM
- Processor : AMD

2.1.2 Software Requirements

- Operating system : WINDOWS 10
- Front end :JAVA 8

  IDE: Eclipse 4.5

  FRAMEWORK: Spring 4.3,Hibernate 4.3
- Back end:PostgreSQL 10

2.2 Software Specification

2.2.1 About Front End

2.2.1.1 java 8

Java is a general-purpose computer-programming language that is concurrent, class-based, object-oriented, and specifically designed to have as few

implementation dependencies as possible. It is intended to let application developers "write once, run anywhere" (WORA),meaning that compiled Java code can run on all platforms that support Java without the need for recompilation Java applications are typically compiled to byte code that can run on any Java virtual machine regardless of computer architecture. As of 2016, Java is one of the most use, particularly for client-server web applications, with a reported 9 million developers. Java was originally developed by James Gosling at Sun Microsystems(which has since been acquired by Oracle Corporation) and released in 1995 as a core component of Sun Microsystems' Java platform. The language derives much of its syntax from C and C++, but it has fewer low-level facilities than either of them.

2.2.1.2 Eclipse

Eclipse is an integrated development environment used in computer programming, and is the most widely used Java IDE. It contains a base workspace and an extensible plug-in system for customizing the environment. Eclipse is written mostly in Java and its primary use is for developing Java applications, but it may also be used to develop applications in other programming languages via plug-ins, including Ada, ABAP, C, C++, C, COBOL, D, Fortran, Haskell, JavaScript, Julia, Lasso, Lua, NATURAL, Perl, PHP, Prolog, Python, R, Ruby (including Ruby on Rails framework), Rust, Scala, Clojure, Groovy, Scheme, and Erlang. It can also be used to develop documents with LaTeX (via a TeXlipse plug-in) and packages for the software Mathematica. Development environments include the Eclipse Java development tools (JDT) for Java and Scala, Eclipse CDT for C/C++, and Eclipse PDT for PHP, among others.

2.2.1.3 Spring Framework

The Spring Framework is an application framework and inversion of control container for the Java platform. The framework's core features can be used by any Java application, but there are extensions for building web appli-

cations on top of the Java EE (Enterprise Edition) platform. Although the framework does not impose any specific programming model, it has become popular in the Java community as an addition to, or even replacement for the Enterprise JavaBeans (EJB) model. The Spring Framework is open source.

Inversion of control container (dependency injection)

Central to the Spring Framework is its inversion of control (IoC) container, which provides a consistent means of configuring and managing Java objects using reflection. The container is responsible for managing object lifecycles of specific objects: creating these objects, calling their initialization methods, and configuring these objects by wiring them together.

Aspect-oriented programming framework

The Spring Framework has its own Aspect-oriented programming (AOP) framework that modularizes cross-cutting concerns in aspects. The motivation for creating a separate AOP framework comes from the belief that it should be possible to provide basic AOP features without too much complexity in either design, implementation, or configuration. The Spring AOP framework also takes full advantage of the Spring container.

Modelviewcontroller framework

The Spring Framework features its own modelviewcontroller (MVC) web application framework. MVC paves the way for cleaner front end code.The DispatcherServlet class is the front controller of the framework and is responsible for delegating control to the various interfaces during the execution phases of an HTTP request.

2.2.1.4 Windows 10

Windows 10 is a personal computer operating system developed and released by Microsoft as part of the Windows NTfamily of operating systems. It was first released on July 29, 2015. Unlike previous versions of Windows,

Microsoft has branded Windows 10 as a ”service” that receives ongoing ”feature updates”. Devices in enterprise environments can receive these updates at a slower pace, or use long-term support milestones that only receive critical updates, such as security patches, over their ten-year lifespan of extended support.

2.2.1.5 Hibernate

Hibernate ORM (Hibernate in short) is an object-relational mapping tool for the Java programming language. It provides a framework for mapping an object-oriented domain model to a relational database. Hibernate handles object-relational impedance mismatch problems by replacing direct, persistent database accesses with high-level object handling functions.

Hibernate's primary feature is mapping from Java classes to database tables, and mapping from Java data types to SQL data types. Hibernate also provides data query and retrieval facilities. It generates SQL calls and relieves the developer from the manual handling and object conversion of the result set.

2.2.1.6 PostgreSQL

PostgreSQL is an object-relational database management system (ORDBMS) based on postgres,version 4.21, developed at the University of California at Berkeley computer science department.postgres pioneered many concepts that only became available in some commercial database systems much later. PostgreSQL is an open-source descendant of this original Berkeley code. It supports a large part of the SQL standard and offers many modern features:

- Complex queries
- Foreign keys
- Triggers
- Updatable views

- Transactional integrity
- Multi version concurrency control

2.3 Security Requirements

2.3.1 Data center

The term data center can be interpreted in a few different ways. First, an organization can run an in-house data center maintained by trained IT employees whose job it is to keep the system up and running. Second, it can refer to an off site storage center that consists of servers and other equipment needed to keep the stored data accessible both virtually and physically.

The New Data Centre will enable various State departments to host their services or applications on a common infrastructure leading to ease of integration and efficient management, ensuring that computing resources and the support connectivity infrastructure (SWAN/NICNET) is adequately and optimally used. These services would be extended to Government agencies, citizens and businesses through the connectivity infrastructure being rolled out such as State Wide Area network (SWAN) and Common Service Centers (CSC). All NIC project are hosted at data centre after security auditing.

2.3.2 Cloud Data center

Cloud services offered by the Data Centre aim to optimise utilization and management of ICT infrastructure and provide agility, reliability, scalability and elasticity in provisioning of resources

- Infrastructure as a service - basic ICT infrastructure offered on demand to users
- Platform as a Service - standardized platforms based on predefined solution stacks are available to the government.

- Platform as a Service - standardized platforms based on predefined solution stacks are available to the government. community. Results in setting up standardized test and development platforms leading to standardization in application development and deployment.

- Virtual Desktop infrastructure - provides users with the desktop which can be accessed anytime, anywhere .

2.3.3 Security Auditing

An Application Security Audit is an assessment of the security risks that are associated with your web applications and client server applications; both those that have external exposure via the internet (such as web shops and customer portals), and those that are part of the internal working of your organisation.

Every vibrant technology marketplace needs an unbiased source of information on best practices as well as an active body advocating open standards. In the Application Security space, one of those groups is the Open Web Application Security Project (or OWASP for short)[11].

As per the guidelines of NIC the web application has to be audit for checking security. Mainly two audits are done.

- Internal Audit: Using the appscan tool the system is tested.Before auditing with the AppScan tool the captcha of the system is disabled and the whole system is run in the frame of the appscan software. the AppScan will check the system against various inputs and the output is verified. Using this tool the top 10 OWASP vulnerabilities are tested and fixed.NIC use IBM Security AppScan.

- External Audit: External audit is the third party.he state government have therefore to ensure cyber security of Kerala felt the necessity for setting up of Computer Emergency Response Team-Kerala (CERT-Kerala or CERT-K) in line with CERT India (CERT-IN). The purpose of CERT-K is to coordinate security efforts and incident response for IT se-

curity problems on a state level in Kerala and to enhance the security of State Government's Communications and Information Infrastructure through proactive action and effective collaboration. CERT-K shall also oversee the implementation of crisis management plan of the Government of Kerala.The Website is tested by the CERT K and if the we application pass all the testes the organization will provide a secure to host certificate for the we application.software application from NIC Kerala can also be externally security audited by the Regional Center for application security which is a competency center for application security set up in NIC Kerala.External Security Audit is a must for hosting the application in any Data Center. External Security Auditing must be done by Government empanelled agencies identified for Security Audit (CERT-IN Empanelled Vendor).External audit can also be handled by STQC, Regional competency center for application security in NIC, are in limitted cases by CERT-K.

2.3.4 Source Code Analysis

Source code analysis is the automated testing of source code for the purpose of debugging a computer program or application before it is distributed or sold. Source code consists of statements created with a text editor or visual programming tool and then saved in a file. The source code is the most permanent form of a program, even though the program may later be modified, improved or upgraded.

Following are the source code analysing tools used in NIC:[8]

- IBM AppScan Source 9.0.34: IBM Security AppScan Source helps organizations develop software code thats secure by design and avoid late-stage vulnerability issues by integrating security testing early in the development cycle.

Source code analysis can be either static or dynamic. In static analysis, debugging is done by examining the code without actually executing the program. This can reveal errors at an early stage in program development,

often eliminating the need for multiple revisions later. After static analysis has been done, dynamic analysis is performed in an effort to uncover more subtle defects or vulnerabilities. Dynamic analysis consists of real-time program testing.

A major advantage of this method is the fact that it does not require developers to make educated guesses at situations likely to produce errors. Other advantages include eliminating unnecessary program components and ensuring that the program under test is compatible with other programs likely to be run concurrently.

CHAPTER 3
SYSTEM ANALYSIS

System Analysis by definition is a process of systematic investigation for the purpose of gathering data, interpreting the facts, diagnosing the problem and using this information to either build a completely new system or to recommend the improvements to the existing system.

A satisfactory system analysis involves the process of examining a business situation with the intent of improving it through better methods and procedures. In its core sense, the analysis phase defines the requirements of the system and the problems which user is trying to solve irrespective of how the requirements would be accomplished.

3.1 Requirements Analysis

Requirements analysis, also called requirements engineering, is the process of determining user expectations for a new or modified product. These features, called requirements, must be quantifiable, relevant and detailed. In software engineering, such requirements are often called functional specifications. Right to Information (RTI) is an Act of the Parliament of India to provide for setting out the practical regime of the right to information for citizens and replaces the erstwhile Freedom of information Act, 2002.Proposed system allow citizen to register and online filing of second appeal. The main purpose of my project is to create a user friendly application which can solve the currently existing issues for filing second appeal.

3.2 Existing System

A Second Appeal must be made within 90 days from the date on which the decision should have been made or from the date a decision was actually received. Currently existing system is manual where all the activity deals with second appeal is offline.

- Steps for Fill in second appeal form, along with the following
  - Original RTI application with its attachments
  - Copy of First appeal with supporting details
  - Receipt of payment of fee using along with proof of additional charges, if any
  - Proof of posting original RTI application and first appeal
  - Decisions of Central Public Information Officer and First Appellate Authority if received
- User can directly give the appeal to state information commission office or Need to send your appeal to the relevant Information Commission in writing.

3.3 Limitations Of Existing System

- The current manual systems suffer from higher rate of inaccuracy, and they are much slower than computerized systems. It waste both money and time
- Offline workflow
- Manual processing is take lots of time
- Work done by both applicant and officers much higher because everything done by human being. Costly
- Not easy to get the status of the appeal need manual efforts
- Traveling expenses

3.4 Proposed System

ONLINE FILING OF RTI SECOND APPEAL AND PROCESSING AT STATE INFORMATION COMMISSION OF KERALA(RASIC)

Proposed system allow citizen to register and online filing of second appeal. RASIC also do the online application processing. RASIC gives the status information about the appeal and Along with these RASIC also provides MIS Reports

List of Modules

1. Citizen Registration RASIC is an online interface to citizens, for filing the appeal the citizen has to register. Citizens can register for RTI account by filling this User Registration Form. After registration they can login to the system[1].

2. Filing Second Appeal User can filing the appeal to the state information commission. On submission of the application, a unique registration number would be issued, which may be referred by the applicant for any references in future.

3. Status tracking Status of the RTI second appeal filed online can be viewed by the applicant and can also get the hearing dates and details

4. Online processing appeal

    - Return the application
    - Give notice of its decision to the complainant and the public authority. The State Information Commission, as the case may be, shall decide the appeal in accordance with such procedure as may be prescribed.
    - If the decision of the State Public Information Officer, as the case may be, against which an appeal is preferred relates to information of a third party, State Information Commission, as the case may

be, shall give a reasonable opportunity of being heard to that third party.

- The workflow includes mainly five types of official users they are Tapal desk, Work desk, Review desk , Secretary , five commissioner and one chief commissioner
- A second appeal first reach to the tapal desk and appeal is checked and forward to work desk. At work desk the appeal either approved and forward to review along with comment or return to the person who file due to incomplete information provided by them. After scrutinize the appeal the reviewer can forward the appeal to secretary. After reaching the appeal at secretary, secretary can approve or reject the appeal.
- If the appeal is approved by secretary the work desk can prepare the notice for the portal user who filed appeal.appeal is then verified by review desk and then reach the secretary. The secretary can approve and send notice to respective person through email.
- After the manual hearing process the appeal can be approved by the commission and final draft can be made

5. MIS Reports
   - Monthly progress reports
   - Pending case reports
   - Monthly report on registered and return cases

3.5 Advantages Of Proposed System

- Processes data and creates reports much faster than manual systems.
- The main difference between current system and new system is the speed.
- Minimizing errors and increasing efficiency

- All transactions can be saved and backed up
- Easy tracking of appeals
- Get updated status and Get MIS Reports
- Easy to get the important dates in case of hearing

3.6 Feasibility Study

A feasibility study is a high-level capsule version of the entire System analysis and Design Process. The study begins by classifying the problem definition. Feasibility is to determine if the project is worth undertaken. Once an acceptance problem definition has been generated, the analyst develops a logical model of the system. A search for alternatives is analyzed carefully. There are 3 parts in feasibility study.

3.6.1 Technical Feasibility

Evaluating the technical feasibility is the trickiest part of a feasibility study. This is because, at this point in time, not too many detailed design of the system, making it difficult to access issues like performance, costs on (on account of the kind of technology to be deployed) etc. A number of issues have to be considered while doing a technical analysis. Understand the different technologies involved in the proposed system before commencing the project we have to be very clear about what are the technologies that are to be required for the development of the new system. Find out whether the organization currently possesses the required technologies. Is the required technology available with the organization? Current technical resources are sufficient for the new system.Unlike traditional applications, web systems are accessible anytime, anywhere via any device with an Internet connection.Now internet,mobiles etc are available in less cost.So that the project is feasible within the limits of current technology.

3.6.2 Operational Feasibility

Proposed project is beneficial only if it can be turned into information systems that will meet the organizations operating requirements. Simply stated, this test of feasibility asks if the system will work when it is developed and installed. Are there major barriers to Implementation? Here are questions that will help test the operational feasibility of a project:

- Is there sufficient support for the project from management from users? If the current system is well liked and used to the extent that persons will not be able to see reasons for change, there may be resistance.
- Are the current business methods acceptable to the user? If they are not, Users may welcome a change that will bring about a more operational and useful systems.
- Have the user been involved in the planning and development of the project? Early involvement reduces the chances of resistance to the system and in general and increases the likelihood of successful project. Since the proposed system was to help reduce the hardships

The web portal help to file the second appeal at Kerala state information commission. The RTI covers all the constitutional authorities, including executive, legislature and judiciary any institution or body established or constituted by an act of Parliament or a state legislature.Now all government department are connected via Kerala State Wide Area Network (KSWAN). This network connects 1500 offices of Government Departments through Wireless and a larger number through Leased Lines and LAN. So that accessing the application and processing of application is easy. By considering these facts RTI second appeal portal is operationally feasible.

3.6.3 Economical Feasibility

Economical feasibility analysis is the most commonly used method for determining the efficiency of a new project. It is also known as cost benefit

analysis also used to conduct Return Of Investment(RIO)analysis, along with tangible and intangible benefits. It helps in identifying profit against investment expected from a project. Cost and time are the most essential factors involved in this field of study.

The conventional file processing is costly and time consuming.The web portal can be access from home and Online filing of second appeal will reduce the traveling expense. No need of extra processing fee.Need minimum cost of processing and producing information.

CHAPTER 4
SYSTEM DESIGN

4.1 Applications Architecture

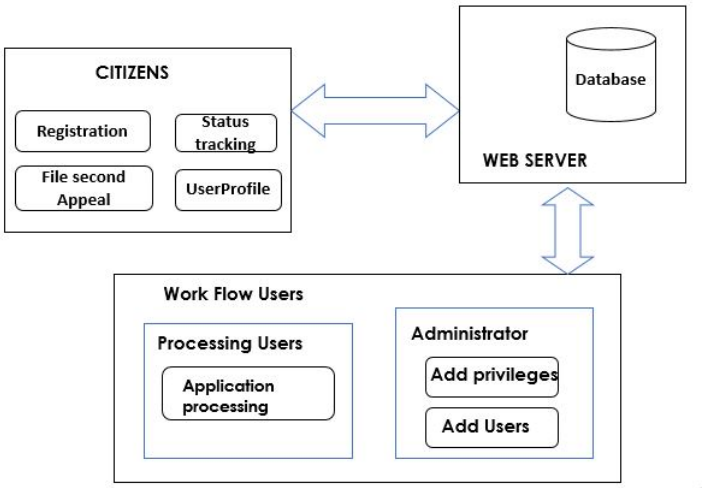


Fig. 4.1: Architecture Diagram

An applications architecture describes the behavior of applications used in a business, focused on how they interact with each other and with users. It is focused on the data consumed and produced by applications rather than their internal structure. In application portfolio management, the applications are usually mapped to business functions and to application.

The system mainly have two types of users one is citizens and other is work flow users. The work flow users they are two type one is administrator who have high privilege and work flow officers.Here the work flow users are tapal desk,work desk,review desk, secretary and commissioner.The final approval is done by commissioner.

4.2 Module Design

List of Modules with description

4.2.1 Citizen Registration

It is an online interface to citizens, for filing the appeal the citizen has to register. Citizens can register for RTI account by filling this User Registration Form. After registration they can login to the system.The system use email OTP for validating the user.After registration the user name and password can be used for further login. The registered Email id can be used to recover the password.

4.2.2 Filing of Second Appeal

User can filing the appeal to the state information commission. On submission of the application, a unique registration number would be issued, which may be referred by the applicant for any references in future.The status of the filed appeal is Application processing and it will be forwarded to tapal desk at State Information Commission

4.2.3 Status tracking

Status of the RTI second appeal filed online can be viewed by the applicant and can also get the hearing dates and details.When the appeal got returned an email will be send to the corresponding user who filed appeal.

4.2.4 Online processing appeal

The work flow users will handle the processing of appeal at State Information Commission.Five work flow users are tapal,Work desk,review desk,Secretary and commissioner

- A second appeal first reach to the tapal desk and it is checked and forward to work desk. At work desk the appeal either approved and forward to review along with comment or return to the person who file

due to incomplete information provided by them. After scrutinize the appeal the reviewer can forward the appeal to secretary. After reaching the appeal at secretary, secretary can approve or reject the appeal.

- If the appeal is approved by secretary the work desk can prepare the notice for the portal user who filed appeal.it is then verified by review desk and then reach the secretary. The secretary can approve and send notice to respective person through email.
- After the manual hearing process the appeal can be approved by the commission and final draft can be made

4.2.5 MIS Reports

- Monthly progress reports
- Pending case reports
- Monthly report on registered and return cases

4.3 Use Case

Representative use case are given below

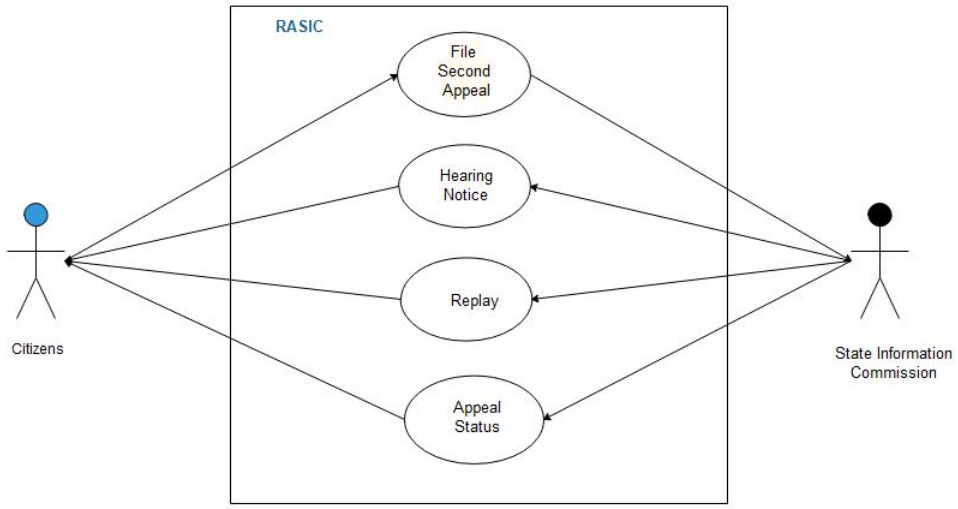


Fig. 4.2: Use case diagram

4.4 User Interfaces Design

The RTI portal has two User interfaces

1 For citizen

2 State government

The public can access information whoever has internet.The modules allow with appropriate user interfaces pertaining to state information commission are provided under the intranet (internal office network)

Loging pages appropriate for the citizen and state information commission are facilitated. The citizen can create own login and credentials along with OTP or email

Necessary dash board with analytic report and MIS are provided with appropriate user interface

Each user is provided with homepage appropriate to the roll of the user in the RTI portal. The menus and functions are organized as per priories .The roll based security is provided in such a way that users of the system will have only those menus and functionality as per their roll in the system.

The various pages of the applications are designed based on the user interface best practices and confirmed GIGW(Guidelines on Indian Government Website) to the extend possible

context sensitive user interfaces, pop up windows and user friendly messages, help messages are provided as appropriate

various functionality are integrated as per the work flow procedures and manage by the stakeholders and users as per the rolls identified by the system.

Essential communication and feed back system shall be facilitated among the various stakeholders of the system utilizing technologies like sms, emails ,dashboards and analysts

4.5 Procedural Design

Software Procedural Design converts and translates structural elements into procedural explanations. SPD starts straight after data design and architectural design. This has now been mostly abandoned mostly due to the rise in preference of Object Oriented Programming and design patterns. A design methodology combines a systematic set of rules for creating a program design with diagramming tools needed to represent it. Procedural designs best used to model programs that have an obvious flow of data from input to output

4.6 Deployment Architecture

Deployment diagram is a structure diagram which shows architecture of the system as deployment (distribution) of software artifacts to deployment targets. Artifacts represent concrete elements in the physical world that are the result of a development process.

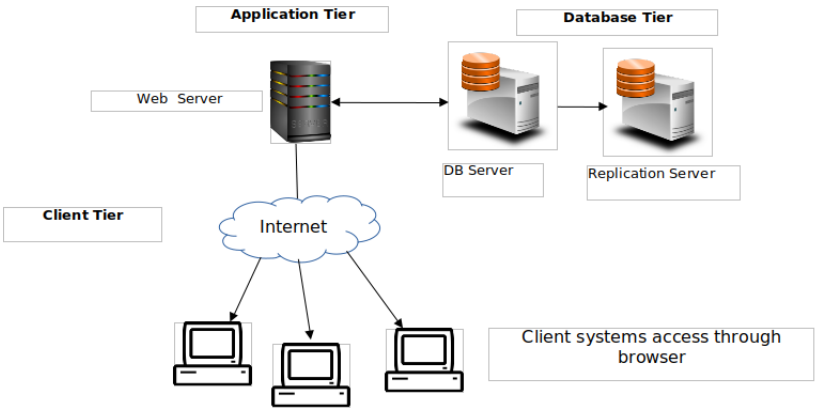


Fig. 4.3: Deployment Architecture

4.6.1 Presentation Layer

The presentation layer is the front end layer in the 3-tier system and consists of the user interface. This user interface is often a graphical one accessible

through a web browser or web-based application and which displays content and information useful to an end user.

For this project ,The presentation tier used web technologies such as HTML5, JavaScript, CSS.

4.6.2 Application Layer

The application layer contains the functional business logic which drives an application's core capabilities. Here the application tier is written in Java.

4.6.3 Data Layer

The data tier comprises of the database/data storage system and data access layer. For my project the data tier made up of PostgreSQL.

4.7 Database Design

Database design is the process of producing a detailed data model of database. This data model contains all the needed logical and physical design choices and physical storage parameters needed to generate a design in a data definition language, which can then be used to create a database. A fully attributed data model contains detailed attributes for each entity. The term database design can be used to describe many different parts of the design of an overall database system. Principally, and most correctly, it can be thought of as the logical design of the base data structures used to store the data. In the relational model these are the tables and views. In an object database the entities and relationships map directly to object classes and named relationships. However, the term database design could also be used to apply to the overall process of designing, not just the base data structures, but also the forms and queries used as part of the overall database application within the database management system (DBMS).

There is a top down approach and bottom up approach for logically designing a database. The ER modeling techniques is the top down, which involves, identifying entity relationship and attributes, drawing ER diagram

and mapping the diagram into tables. Normalization is the Bottom up approach.

4.7.1 Normalization

Normalization is the process of organizing the columns (attributes) and tables (relations) of a relational database to reduce data redundancy and improve data integrity. Normalization is also the process of simplifying the design of a database so that it achieves the optimal structure composed of atomic elements.

First normal form

first normal form (1nf) is a property of a relation in a relational database. A relation is in first normal form if and only if the domain of each attribute contains only atomic (indivisible) values, and the value of each attribute contains only a single value from that domain. First normal form enforces these criteria:

- Eliminate repeating groups in individual tables.
- Create a separate table for each set of related data
- Identify each set of related data with a primary key

Second normal form

a relation is in 2nf if it is in 1nf and no non-prime attribute is dependent on any proper subset of any candidate key of the relation. A non-prime attribute of a relation is an attribute that is not a part of any candidate key of the relation. Put simply, a relation is in 2nf if it is in 1nf and every non-prime attribute of the relation is dependent on the whole of every candidate key.

Third Normal form

A table design is said to be in 3NF if both the following conditions hold:
1. Table must be in 2NF

2. Transitive functional dependency of non-prime attribute on any super key should be removed.

Boyce Codd normal form (BCNF)

It is an advance version of 3NF thats why it is also referred as 3.5NF. BCNF is stricter than 3NF. A table complies with BCNF if it is in 3NF and for every functional dependency X- >Y, X should be the super key of the table.

4.7.2 Table Design

Fields name	**Data type**	**Size**	**constraints**
apl_status_id	smallint		PRIMARY KEY
status_desc_eng	character varying	100	NOT NULL
status_desc_mal	character varying	100	NOT NULL
updatedon	timestamp		NOT NULL
updatedby	integer		NOT NULL

Table 4.1: apl_ status_ master

Fields name	**Data type**	**Size**	**constraints**
authorization_id	smallint		PRIMARY KEY
authorization_name_eng	character varying	50	NOT NULL
authorization_name_mal	character varying	50	NOT NULL
updatedon	Timestamp without time zone		NOT NULL
updatedby	integer		FOREIGN KEY
dept_id	integer		NOT NULL

Table 4.2: Authorizations

Feilds name	Data type	Size	Constraints
service_id	serial		PRIMARY KEY
authorization_id	smallint		FOREIGN KEY
menu_id	smallint		NOT NULL
updatedon	timestamp without time zone		NOT NULL
updatedby	integer		FOREIGN KEY

Table 4.3: service_roles

Feilds name	Data type	Size	constraints
office_level_id	integer	100	PRIMARY KEY FOREIGN KEY
office_level_name	character varying	50	

Table 4.4: office_level

Feilds name	Data type	Size	Constraints
dept_id	smallint		PRIMARY KEY
dept_name_eng	character varying	50	NOT NULL
dept_name_mal	character varying	50	NOT NULL
updatedon	Timestamp without time zone		NOT NULL
updatedby	integer		FOREIGN KEY

Table 4.5: Department

Feilds name	Data Type	Size	Constraints
district_id	smallint		PRIMARY KEY
state_id	character varying	50	FOREIGN KEY
district_name_eng	character varying	50	NOT NULL
district_name_mal	character varying	50	NOT NULL
updatedon	Timestamp without time zone		NOT NULL
updatedby	integer		FOREIGN KEY

Table 4.6: District

Feilds name	Data type	Size	constraints
id	serial		PRIMARY KEY
user_id	character varying	50	FOREIGN KEY
attempt_time	character varying	50	
ip_address	character varying		NOT NULL
mac_address	character varying		FOREIGN KEY

Table 4.7: failed_attempts

Feilds name	Data type	Size	Constraints
cert_type_id	smallint		PRIMARY KEY
cert_name_eng	character varying	50	NOT NULL
cert_name_mal	character varying	50	NOT NULL
cert_validity_type	character varying	50	NOT NULL
validity_month	smallint	6	FOREIGN KEY
issue_dept	character varying	50	FOREIGN KEY
gen_form_name_eng	character varying	50	
gen_form_name_mal	character varying	50	
specific_form_name_eng	character varying	50	
specific_form_name_mal	character varying	50	
backend_type	boolean		
ed_ws_url_process_status	character varying	100	
ed_ws_url_verify_status	character varying	100	
ed_ws_url_approve_status	character varying	100	
remote_ws_url	timestamp		
updatedon	integer		NOT NULL
Updated_by	integer		FOREIGN KEY
service_code	character varying	50	

Table 4.8: Cert_type

Feilds name	Data type	Size	constraints
user_id	serial		PRIMARY KEY
login_name	character varying	20	UNIQUE, NOT NULL
upassword	character varying	100	NOT NULL
user_type	smallint		
username	smallint	99	NOT NULL
emp_code	character varying	6	NOT NULL
dept_id	character varying	3	FOREIGN KEY
office_id	character varying	2	FOREIGN KEY
bio_enabled	character varying	2	DEFAULT false
ds_enabled	character varying	2	DEFAULT false
allow_login	boolean	2	DEFAULT true
enroll_date	date		
fail_atmpts_no	smallint		DEFAULT 0
updatedon	Timestamp without time zone		NOT NULL
pwd_changed_date	timestamp without time zone		
security_question	character varying	100	
security_answer	character varying	50	
updatedby	integer	50	FOREIGN KEY
initial_login	boolean		
user_designation	character varying	50	NOT NULL
passwordold	character varying	100	
passwordolder	character varying	100	
pwd_reset	boolean		

Table 4.9: edUser

Feilds name	Data type	Size	constraints
user_id	integer		PRIMARY KEY FOREIGN KEY
email_id	character varying	50	
mobile_no	character varying	50	FOREIGN KEY
updatedon	timestamp without time zone		
updatedby	integer		FOREIGN KEY

Table 4.10: eduser_addl_info

Feilds name	Data type	Size	constraints
lan_id	integer		PRIMARY KEY
lan_name_eng	character varying	50	NOT NULL
lan_name_mal	character varying	50	NOT NULL
updatedon	timestamp without time zone		NOT NULL
updatedby	integer		FOREIGN KEY

Table 4.11: language

Feilds name	Data type	Size	constraints
question_id	serial		PRIMARY KEY
question	character varying	100	
updatedon	date		NOT NULL
updatedby	integer		NOT NULL

Table 4.12: security_question

Feilds name	Data type	Size	constraints
menu_id	smallint		PRIMARY KEY
menu_caption	character varying	50	NOT NULL
menu_path	character varying	50	
menu_path_mal	character varying	100	
menu_order	smallint		
menu_type	character varying	30	
parent_menu_id	smallint		
updatedon	timestamp without time zone		NOT NULL
updatedby	integer		FOREIGN KEY
dept_id	character varying	8	

Table 4.13: menu

Feilds name	Data type	Size	Constraints
dept_id	character varying	3	PRIMARY KEY
office_id	character varying	8	FOREIGN KEY
user_id	integer		FOREIGN KEY
authorization_id	smallint		FOREIGN KEY
cert_type_id	smallint		FOREIGN KEY
from_date	date		
to_date	date		
updatedon	timestamp without time zone		NOT NULL
updatedby	integer		FOREIGN KEY
actual_charge	character varying	1	
active	boolean		

Table 4.14: privileges

Feilds name	Data type	Size	constraints
pwd_pol_id	integer		PRIMARY KEY
pwd_wef	date		NOT NULL
pwd_min_len	smallint		NOT NULL
pwd_max_len	smallint		NOT NULL
max_attempts	smallint		NOT NULL
pwd_validity	smallint		NOT NULL
updatedon	timestamp without time zone		NOT NULL
updatedby	integer		FOREIGN KEY

Table 4.15: password_policy

Feilds name	Data type	Size	constraints
draft_id	serial		PRIMARY KEY
rti_hand_id	integer		
draft	character varying	200000	
updated_on	timestamp without time zone		
created_by	integer		FOREIGN KEY
review_by	integer		FOREIGN KEY
review_date	date		
secretary	integer		FOREIGN KEY
approve_date	date		
comm_date	date		

Table 4.16: Draft

Feilds name	Data type	Size	constraints
notice_id	serial		PRIMARY KEY
nt_heading	character varying	100	
nt_description	character varying	500	
rti_hand_id	integer		FOREIGN KEY
nt_status	integer		
from_seat	integer		FOREIGN KEY
to_seat	integer		FOREIGN KEY
updated_on	timestamp without time zone		
hr_mode	character varying	50	
venue	character varying	200	
nt_entered_on	date		
hr_date	date		
hr_time	timestamp without time zone		
doc	bytea		
doc_path	character varying	1000	
doc_updation	timestamp without zone		

Table 4.17: Notice

Feilds name	Data type	Size	constraints
id	integer		PRIMARY KEY
p_user_id	date		FOREIGN KEY
attempt_time	timestamp without time zone		NOT NULL
ip_address	character varying	50	NOT NULL
mac_address	character varying	50	

Table 4.18: portal_failed_attempts

Feilds name	Data type	Size	Constraints
reason_id	character varying	50	PRIMARY KEY
reason_name_eng	character varying	100	
reason_name_mal	character varying	100	
exclude_status	character varying	100	
updated_by	integer		
updated_on	timestamp without time zone		

Table 4.19: rti_appeal_reasons

Feilds name	Data type	Size	constraints
Cmnt_id	serial		PRIMARY KEY
rti_hand_id	integer		
from_seat	integer		
to_seat	integer		
updated_on	timestamp without time zone		
comment	character varying	500	
flag	boolean		

Table 4.20: Sa_Comment

Feilds name	Data type	Size	constraints
office_id	serial		PRIMARY KEY
dept_id	character varying	20	FOREIGN KEY
district_id	character varying	100	FOREIGN KEY
taluk_id	smallint		FOREIGN KEY
jurisdiction_id	smallint	99	FOREIGN KEY
office_name_eng	character varying	6	
office_address_eng	character varying	3	
office_name_mal	character varying	2	
office_address_mal	character varying	2	
reporting_office_id	character varying	2	
officer_name	boolean	2	
reporting_office_dept	smallint		FOREIGN KEY
lb_id	character varying	100	FOREIGN KEY
block_id	character varying	50	FOREIGN KEY
updatedby	integer	50	FOREIGN KEY
updatedon	boolean		
office_level_id	character varying	50	FOREIGN KEY
center_code	character varying	100	
off_code	character varying	100	

Table 4.21: office_csc_master

Feilds name	Data type	Size	constraints
p_user_id	integer		PRIMARY KEY (p_user_id logout_time)
timestamp with time zone	timestamp with time zone		FOREIGN KEY
mac_address	character varying	50	NOT NULL
logout_time	character varying	50	NOT NULL
logout_type	character varying	30	

Table 4.22: portal_login_logs

Feilds name	Data type	Size	constraints
pwd_pol_id	smallint		PRIMARY KEY
pwd_wef	date		NOT NULL
pwd_min_len	smallint		NOT NULL
pwd_max_len	smallint		NOT NULL
max_attempts	smallint		NOT NULL
pwd_validity	smallint		NOT NULL
updatedon	timestamp without time zone		NOT NULL
updatedby	integer		FOREIGN KEY

Table 4.23: portal_password_policy

Feilds name	Data type	Size	Constraints
p_user_id	serial		PRIMARY KEY
p_user_loginname	character varying	50	NOT NULL UNIQUE
ppassword	character varying	200	NOT NULL
pusername	character varying	99	NOT NULL
deny_login	boolean		DEFAULT FALSE
fail_atmpts_no	smallint		DEFAULT 0
security_answer	character varying	50	
updatedon	timestamp without time zone		
gender	character	1	
dob	date		
house_no_name	character varying	50	
locality	character varying	20	
post_office	character varying		PRIMARY KEY
pin_code	character varying		
mobile_no	character varying		
email_id	character varying		NOT NULL
security_qns_id	integer		FOREIGN KEY
uid_no	numeric		

Table 4.24: portal_user

Feilds name	Data type	Size	Constraints
rti_hand_id	serial		PRIMARY KEY
appeal_type	character varying	3	NOT NULL
first_app_num	character varying	10	NOT NULL
first_district_id	character varying	15	FOREIGN KEY
first_dept_id	character varying	3	FOREIGN KEY
first_office_id	character varying	8	FOREIGN KEY
rti_district_id	character varying	15	FOREIGN KEY
rti_office_id	character varying	3	FOREIGN KEY
rti_dept_id	character varying	8	FOREIGN KEY
reason_id	integer		FOREIGN KEY
second_description	character varying	20000	NOT NULL
first_appeal_date	date	50	NOT NULL
second_appeal_date	date		NOT NULL
puser_id	character varying	10	FOREIGN KEY
rti_req_num	character varying	15	NOT NULL
rti_date	date		NOT NULL
language	integer		FOREIGN KEY
appeal_status	smallserial		FOREIGN KEY

Table 4.25: second_appeal

4.7.3 Input Design

Input design is the link between information system and the user. It comprises of determining set of inputs, validates the data, minimizes the data entries and thereby provides multiuser facilities. The input is designed in such a way that it provides security and ease of use with retaining privacy. Input Design is the process of converting a user-oriented description of the input into a computer-based system. This design is important to avoid errors in the data input process and show the correct direction to the management for getting correct information from the computerized system. It is achieved by creating user-friendly screens for the data entry to handle large volume of data. The goal of designing input is to make data entry easier and to be free from errors. Once identified, the appropriate input media are selected for processing. All the input data are validated and if any data violates any conditions, the user is warned by a message. If the data satisfies all the conditions, it is transferred to the appropriate tables in the database.

4.7.4 Output Design

A quality output is one, which meets the requirements of the end user and presents the information clearly. In any system results of processing are communicated to the users and to other system through outputs. In output design it is determined how the information is to be displaced for immediate need and also the hard copy output. It is the most important and direct source information to the user. Efficient and intelligent output design improves the system's relationship to help user decision-making. Designing computer output should proceed in an organized, well thought out manner; the right output must be developed while ensuring that each output element is designed so that people will find the system can use easily and effectively.

4.8 Data Flow Diagram

A data flow diagram is a graphical representation of the flow of data through an information system, modeling its process aspects. A DFD is often used as a preliminary step to create an overview of the system, which can later be elaborated. DFDs can also be used for the visualization of data processing (Structured Design). A DFD shows what kind of information will be input to and output from the system, where the data will come from and go to, and where the data will be stored. It does not show information about the timing of process or information about whether processes will operate in sequence or in parallel.

4.8.1 List of Symbols

For constructing a DFD there are some symbols and notations to be considered. Primitive symbols used for constructing DFDs is,

Process Symbol -Circle

An activity that changes or transforms data flows. Since they transform incoming data to outgoing data, all processes must have inputs and outputs on a DFD. This symbol is given a simple name based on its function, such as login, rather than being labeled process on a diagram.

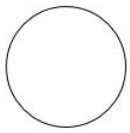


Fig. 4.4: Process Symbol

External Entity Symbol -Rectangle

Also known as actors, sources or sinks, and terminators, external entities produce and consume data that flows between the entity and the system being diagrammed. These data flows are the inputs and outputs of the

DFD. Since they are external to the system being analyzed, these entities are typically placed at the boundaries of the diagram. They can represent another system or indicate a subsystem.



Fig. 4.5: External Entity Symbol

Dataflow Symbol -Arrows

Movement of data between external entities, processes and data stores is represented with an arrow symbol, which indicates the direction of flow. This data could be electronic, written or verbal. Input and output data flows are labeled based on the type of data or its associated process or data store, and this name is written alongside the arrow.



Fig. 4.6: Dataflow Symbol

Data Store Symbol -Open Rectangle

A data store does not generate any operations but simply holds data for later access. Data stores could consist of files held long term or a batch of documents stored briefly while they wait to be processed. Input flows to a data store include information or operations that change the stored data. Output flows would be data retrieved from the store.



Fig. 4.7: External Entity Symbol

Data Flow Diagram



Fig. 4.8: LEVEL 0 DFD

The level one users of the system is portal user and official users ,they can login to the system using user name and password

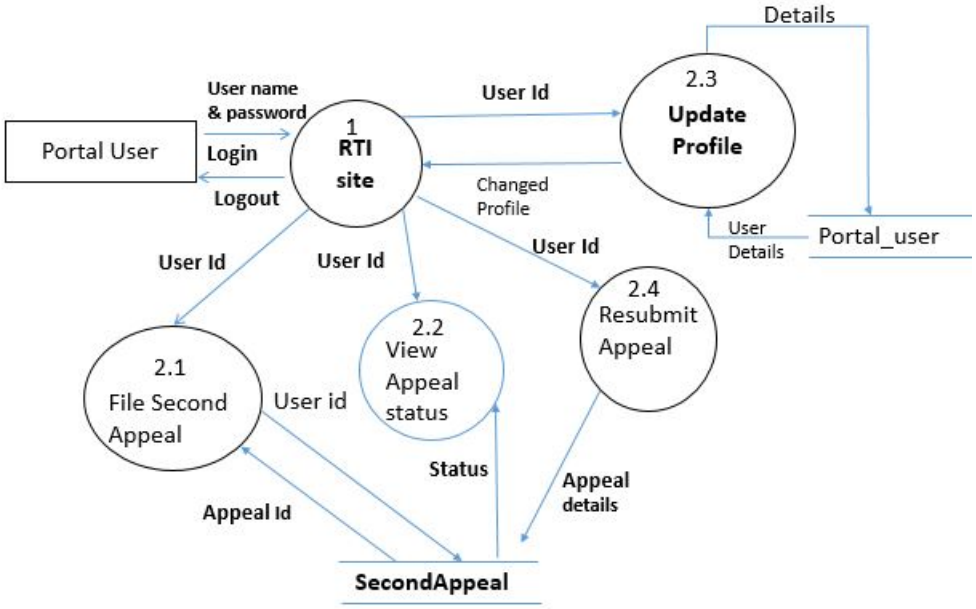


Fig. 4.9: LEVEL 1 DFD- Portal User

After login to the portal,the user can file the second appeal, view appeal status,update profile and can resubmit the appeal with edited details.

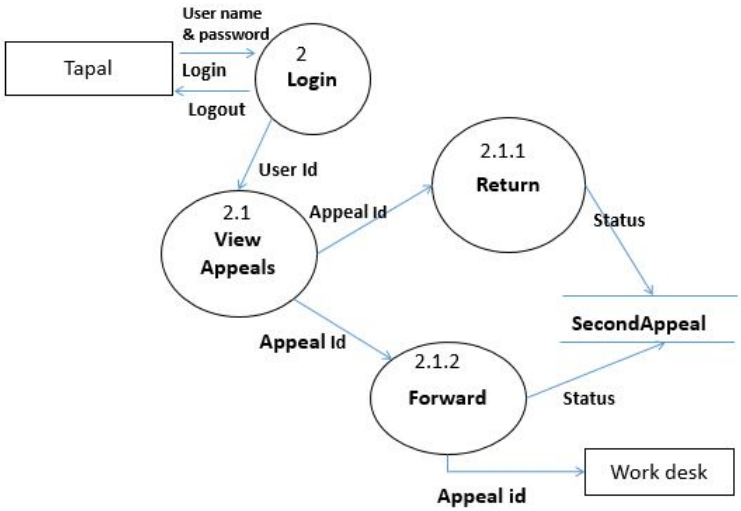


Fig. 4.10: LEVEL 1 DFD- Tapal

The tapal user can forward the appeal for work desk

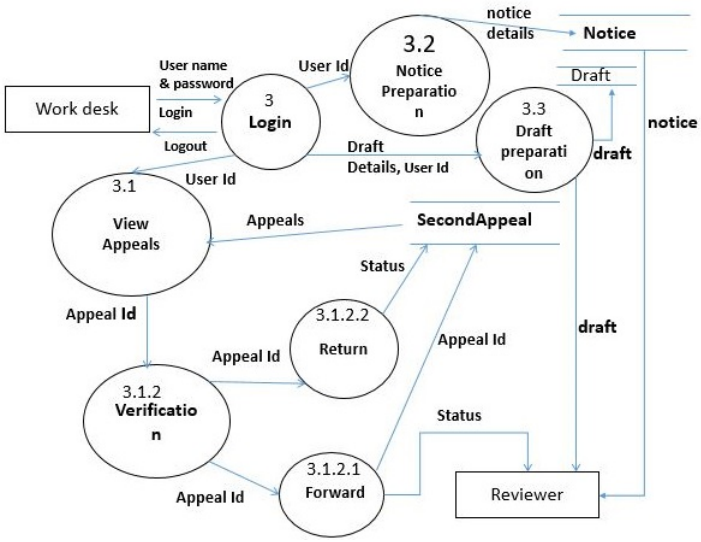


Fig. 4.11: LEVEL 1 DFD- Work desk

After login the work desk can view appeals forwarded by the tapal desk, or can reject the appeal.The work desk can create notice and draft and forward to review desk.

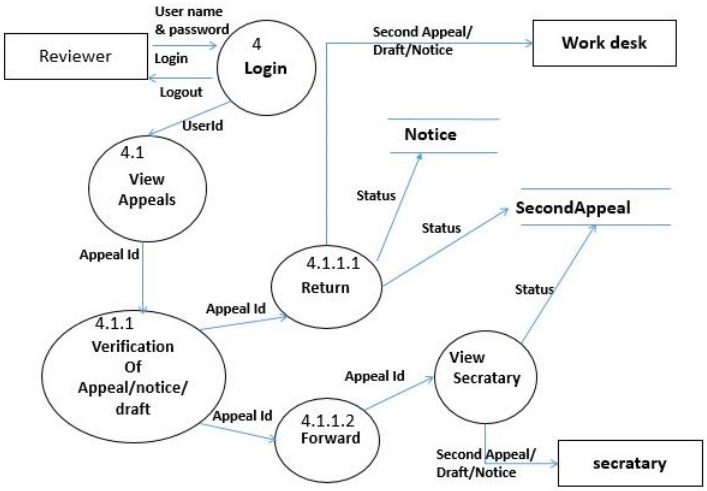


Fig. 4.12: LEVEL 1 DFD- Review desk

After login the review desk can verify the appeals forwarded by the work desk, or can reject the appeal.The work desk can verify notice and draft and forward to secretary desk.

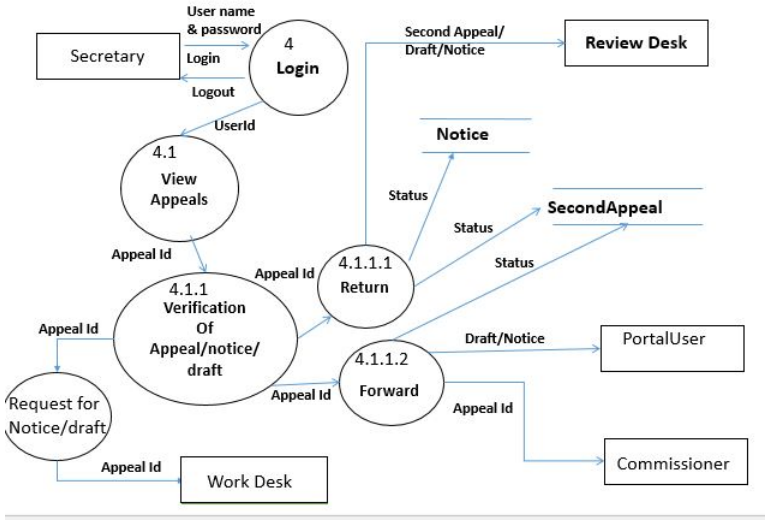


Fig. 4.13: LEVEL 1 DFD-Secretary

After login the review desk can verify the appeals forwarded by the review desk, or can reject the appeal.the secretary can verify notice and draft and forward to the applicant.

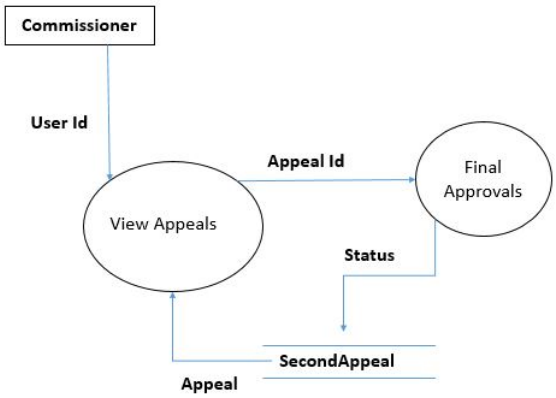


Fig. 4.14: LEVEL 1 DFD- Commissioner

The commissioner has the person who has the power to approve the appeal.

4.9 ER Diagram

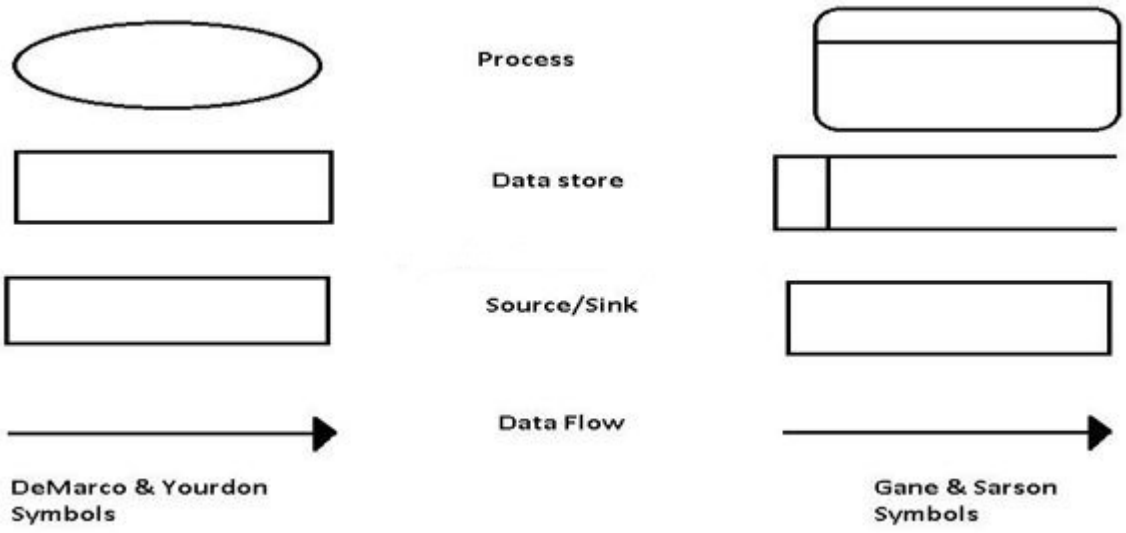


Fig. 4.15: ER Diagram Symbols

An entityrelationship model (ER model) describes inter-related things of interest in a specific domain of knowledge. An ER model is composed of entity types (which classify the things of interest) and specifies relationships that can exist between instances of those entity types.

4.9.1 Common ER-Diagram Symbols

Entities, which are represented by rectangles. An entity is an object or concept about which you want to store information.



Fig. 4.16: Entity Symbol

Actions, which are represented by diamond shapes, show how two entities share information in the databases.



Fig. 4.17: Relationship Symbol

Attributes, which are represented by ovals. A key attribute is the unique, distinguishing characteristic of the entity.



Fig. 4.18: Attribute Symbol

Connecting lines, solid lines that connect attributes to show the relationships of entities in the diagram.

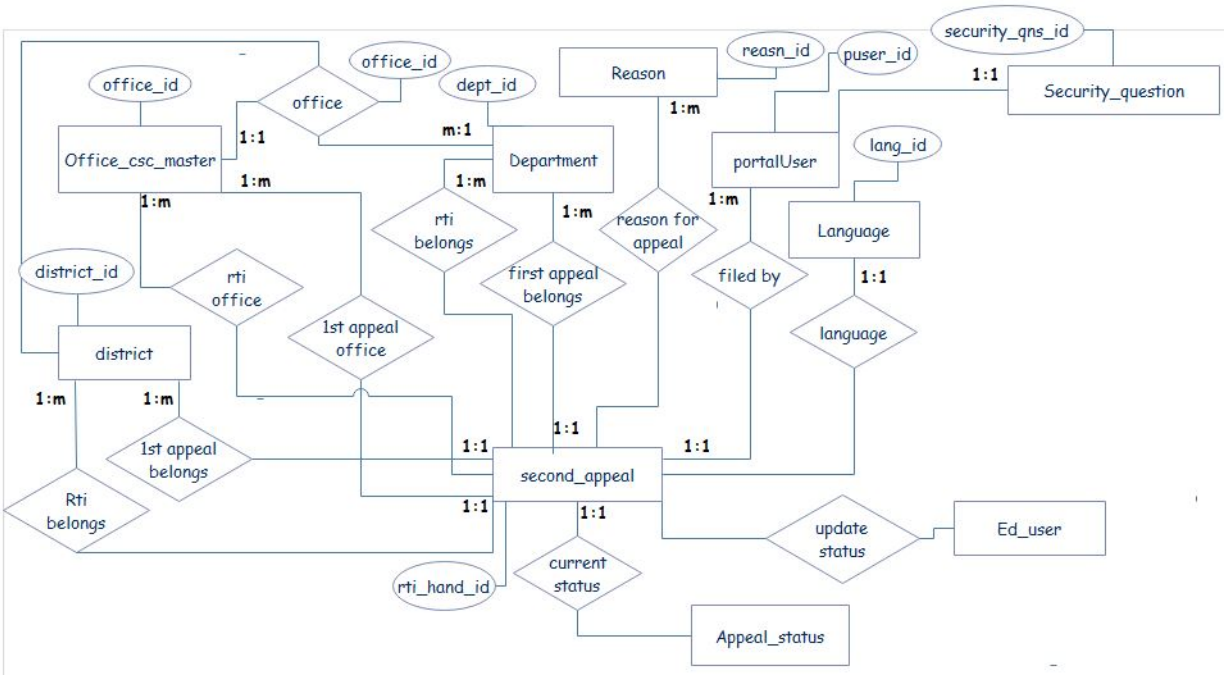


Fig. 4.19: ER diagram

CHAPTER 5
SYSTEM IMPLEMENTATION AND TESTING

5.1 System Implementation

Implementation is the process of bringing a newly developed system or devised into operational one. It is the practical job of putting a theoretical design into practice. The new system and its components are to be tested in a structured and planned manner. The implementation stage of a project is often very complex and time consuming and many more people are involved in the earlier stages. This involves careful planning, investigation of the current system and constraints of implementation, creating compatible , installing hardware, training the operating stages in the changeover procedures before the system is setup and running. A critical factor in the conversion is not disrupting the functioning of the organization. System implementation is the process of defining how the information system should be built (physical system design), ensuring that the information system is operational and used, ensuring that the information system meets quality standard. In computer science, an implementation is a realization of a technical specification or algorithm as a program, software component, or other computer system through computer programming and deployment. The implementation phase of a project covers the period from the acceptance of the tested design to its satisfactory operations, supported by the appropriate user and operator's manual. It is a major operation across the whole organizational structure and require a great deal of planning. The implementation plan involves the following:

1. Testing to confirm effectiveness
2. Detection and correction of errors
3. Making necessary changes so as to satisfy the requirements

5.2 System Testing

Software testing is evaluation of the software against requirements gathered from users and system specifications. Testing is conducted at the phase level in software development life cycle or at module level in program code. Software testing comprises of Validation and Verification

5.2.1 Validation

Validation is process of examining whether or not the software satisfies the user requirements. It is carried out at the end of the Software development life cycle. If the software matches requirements for which it was made, it is validated.

The validations for client side and server side is implemented in the system.At client side validation for null values ,The illegal data and date validation are done for web portal.The validations for data type also checked. At server side the validations of data and validity of date and data type are implemented.

The following are the some of major validation incorporated in the system.

1. The RTI petition filed by the citizen is forwarded to right person in SIC.
2. The person files petition get sms to right petitioner about the right petition.
3. The filed appeal strictly flow the work flow

5.2.2 Verification

Verification is the process of confirming if the software is meeting the business requirements, and is developed adhering to the proper specifications

and methodologies. Tests can be conducted based on two approaches:

1. Functionality testing
2. Implementation testing

Software requirement specification are verified against the user interface .The modular design , business requirements, logic implemented , functionalities of the system ,procedures of the system are verified and expected results are obtained

5.2.3 Black box testing

It is carried out to test functionality of the program. It is also called 'Behavioral' testing. The tester in this case, has a set of input values and respective desired results. On providing input, if the output matches with the desired results, the program is tested 'ok', and problematic otherwise

Both valid and invalid data are used for testing and got the desired result.

5.2.4 White box testing

It is a method of testing software that tests internal structures or workings of an application, as opposed to its functionality. In white-box testing an internal perspective of the system, as well as programming skills, are used to design test cases. The tester chooses inputs to exercise paths through the code and determine the appropriate outputs. This is analogous to testing nodes in a circuit, e.g. in-circuit testing (ICT). White-box testing can be applied at the unit, integration and system levels of the software testing process. Although traditional testers tended to think of white-box testing as being done at the unit level, it is used for integration and system testing more frequently today. It can test paths within a unit, paths between units during integration, and between subsystems during a system level test. Though this method of test design can uncover many errors or problems, it has the potential to miss unimplemented parts of the specification or missing requirements.

white box testing is performed successfully.

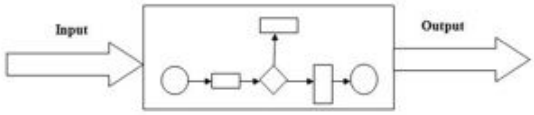


Fig. 5.1: White box testing

Testing levels

Testing itself may be defined at various levels of software development cycle. The testing process runs parallel to software development. Before jumping on the next stage, a stage is tested, validated and verified. Testing separately is done just to make sure that there are no hidden bugs or issues left in the software. Software is tested on various levels:

5.2.5 Unit testing

While coding, the programmer performs some tests on that unit of program to know if it is error free. Testing is performed under white-box testing approach. Unit testing helps developers decide that individual units of the program are working as per requirement and are error free.

5.2.6 Integration testing

Even if the units of software are working individually, there is a need to test if the units if integrated together would also work without errors. For example, argument passing and data updating etc.
Integration testing is performed in the system.Some example testing done in the system are
1. If a citizen file RTI appeal then appeal should appear under the application processing state to relevant officer in tapal
2. If the result of petition is prepared then the result must shown to corresponding user page of RTI portal

3. Menu appear after checking priority of the user

5.2.7 System testing

The software is compiled as product and then it is tested as a whole. This can be accomplished using one or more of the following tests:

5.2.8 Functionality testing

This test proves how efficient the software is. It tests the effectiveness and average time taken by the software to do desired task. Performance testing is done by means of load testing and stress testing. The RTI appeal is verified and take decision with in the number of days as per the RTI act.

5.2.9 Open Web Application Security Project

The Open Web Application Security Project (OWASP) is a 501(c)(3) worldwide not-for-profit charitable organization focused on improving the security of software.

Our mission is to make software security visible, so that individuals and organizations are able to make informed decisions. OWASP is in a unique position to provide impartial, practical information about AppSec to individuals, corporations, universities, government agencies and other organizations worldwide. Operating as a community of like-minded professionals, OWASP issues software tools and knowledge-based documentation on application security.

The software has to be audited against known stranded vulnerabilities that may effect application. Typically there are 10 vulnerabilities defined by OWSAP.The following are the top 10 vulnerabilities defined by OWSAP 2017

5.2.10 OWASP-Top 10 Vulnerabilities 2017

1. Injection :Injection flaws, such as SQL injection, LDAP injection, and CRLF injection, occur when an attacker sends untrusted data to an

interpreter that is executed as a command without proper authorization.Application security testing can easily detect injection flaws. Developers should use parameterized queries when coding to prevent injection flaws.

2. Broken Authentication and Session Management :Incorrectly configured user and session authentication could allow attackers to compromise passwords, keys, or session tokens, or take control of users accounts to assume their identities. Multi-factor authentication, such as FIDO or dedicated apps, reduces the risk of compromised accounts.

3. Sensitive Data Exposure: Applications and APIs that dont properly protect sensitive data such as financial data, usernames and passwords, or health information, could enable attackers to access such information to commit fraud or steal identities.Encryption of data at rest and in transit can help you comply with data protection regulations.

4. XML External Entity: Poorly configured XML processors evaluate external entity references within XML documents. Attackers can use external entities for attacks including remote code execution, and to disclose internal files and SMB file shares.Static application security testing (SAST) can discover this issue by inspecting dependencies and configuration.

5. Broken Access Control :Improperly configured or missing restrictions on authenticated users allow them to access unauthorized functionality or data, such as accessing other user's accounts, viewing sensitive documents, and modifying data and access rights.Penetration testing is essential for detecting non-functional access controls; other testing methods only detect where access controls are missing.

6. This risk refers to improper implementation of controls intended to keep application data safe, such as misconfiguration of security headers, error messages containing sensitive information (information leak-

age), and not patching or upgrading systems, frameworks, and components.Dynamic application security testing (DAST) can detect misconfigurations, such as leaky APIs.

7. Cross-Site Scripting :Cross-site scripting (XSS) flaws give attackers the capability to inject client-side scripts into the application, for example, to redirect users to malicious websites.Developer training complements security testing to help programmers prevent cross-site scripting with best coding best practices, such as encoding data and input validation.

8. Insecure deserialization flaws can enable an attacker to execute code in the application remotely, tamper or delete serialized (written to disk) objects, conduct injection attacks, and elevate privileges. Application security tools can detect deserialization flaws but penetration testing is frequently needed to validate the problem.

9. Using Components With Known Vulnerabilities: Developers frequently dont know which open source and third-party components are in their applications, making it difficult to update components when new vulnerabilities are discovered. Attackers can exploit an insecure component to take over the server or steal sensitive data. Software composition analysis conducted at the same time as static analysis can identify insecure versions of components.

10. Insufficient Logging and Monitoring :The time to detect a breach is frequently measured in weeks or months. Insufficient logging and ineffective integration with security incident response systems allow attackers to pivot to other systems and maintain persistent threats. Think like an attacker and use pen testing to find out if you have sufficient monitoring; examine your logs after pen testing.

All these vulnerabilities are checked and fixed in internal auditing.

At NIC every application are tested using the following tools

1. Source code analysis

2. AppScan audit tools

As per the completion of internal audit the software given to external audit.In NIC the external audit is done at the data centers .The Data Centre will enable various State departments to host their services or applications on a common infrastructure leading to ease of integration and efficient management, ensuring that computing resources and the support connectivity infrastructure (SWAN/NICNET) is adequately and optimally used.

5.3 Test Case

A TEST CASE is a set of conditions or variables under which a tester will determine whether a system under test satisfies requirements or works correctly. The process of developing test cases can also help find problems in the requirements or design of an application.

S.no	TestCase ID	Module Name	Description	Expected Output	Actual Output	Result
1	User_01	Citizen Registration and login	Only the user registered RTI portal can login. Hence he is a authorized user	Registered user can login and unauthorized user do not login	Valid user login successfully	Pass
2	Office_01	Online processing	Only the registered Tapal user in the SIC can login. Hence he is a authorized user	Registered user can login and unauthorized user do not login	Valid officers login successfully	Pass
3	Change pwd_03	Change password	We can change the password of Portal User	Password can change by the user itself.	display the individual user details	Pass

S.no	TestCase ID	Module Name	Description	Expected Output	Actual Output	Result
4	Appeals 04	Online processing	List out Filed appeals	Display the all details of corresponding user	Display the all details of corresponding user	Pass
5	Status_05	Online processing	For each state in workflow of appeal verification produce different status to appeal	Change status accordingly	Change status accordingly	Pass
6	Email_06	Online processing	A notification mail send to user when the appeal is returned	Send notification of returned appeal to mail of user	Send notification of returned appeal to mail corresponding user	pass
7	Tracking status	Status tracking	The user can track the status of his filed appeal	The appellant can see their current status by login to the system	The appellant can see their current status by login to the system	pass

Table 5.1: Test cases

Test cases are tested and system is accepted

CHAPTER 6
CONCLUSION

Anybody who is unhappy or dissatisfied with the decision of the RTI Appellate Authority can file Second Appeal to the Information Commission at the respective States. This portal allow RTI second appeal can be filed by Indian Citizens for all Ministries/Departments and other Public Authorities of Kerala state Government. Easy way to file the second appeal to state information commission say to track the status. Online processing of received appeal at SIC also will be addressed by the system. Easy tracking of second appeal status and get notification of hearing date, draft etc. make this portal interactive to common people.

CHAPTER 7

SCOPE FOR THE FUTURE ENHANCEMENT

The effectiveness and efficiency of the proposed system can be improved and enhanced in the way of capacity, Security and robustness. Proposed web application to handle second appeal of RTI application at SIC Kerala.In future the system can add payment gateway integration for online payment of penalty imposed on officers for delaying on not giving of information. This web portal will be integrate with the RTI portal of the govt. of Kerala under the eDistrict platform.

CHAPTER 8

SCREENSHOTS

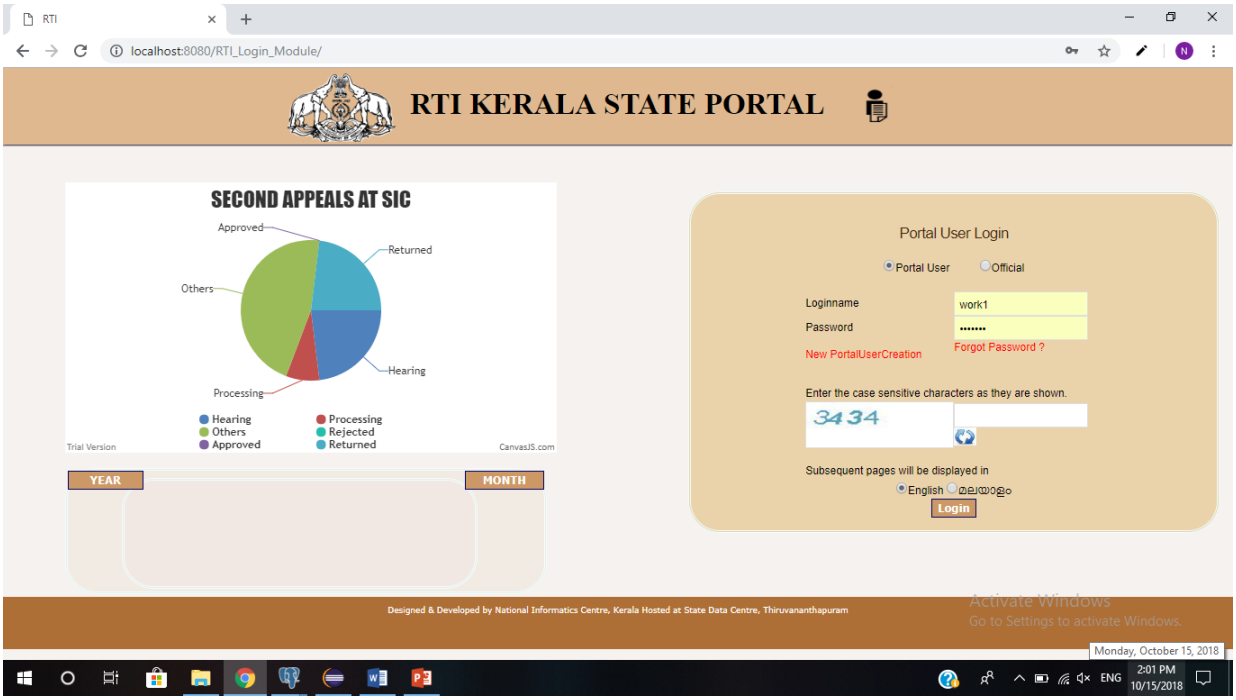


Fig. 8.1: Homepage

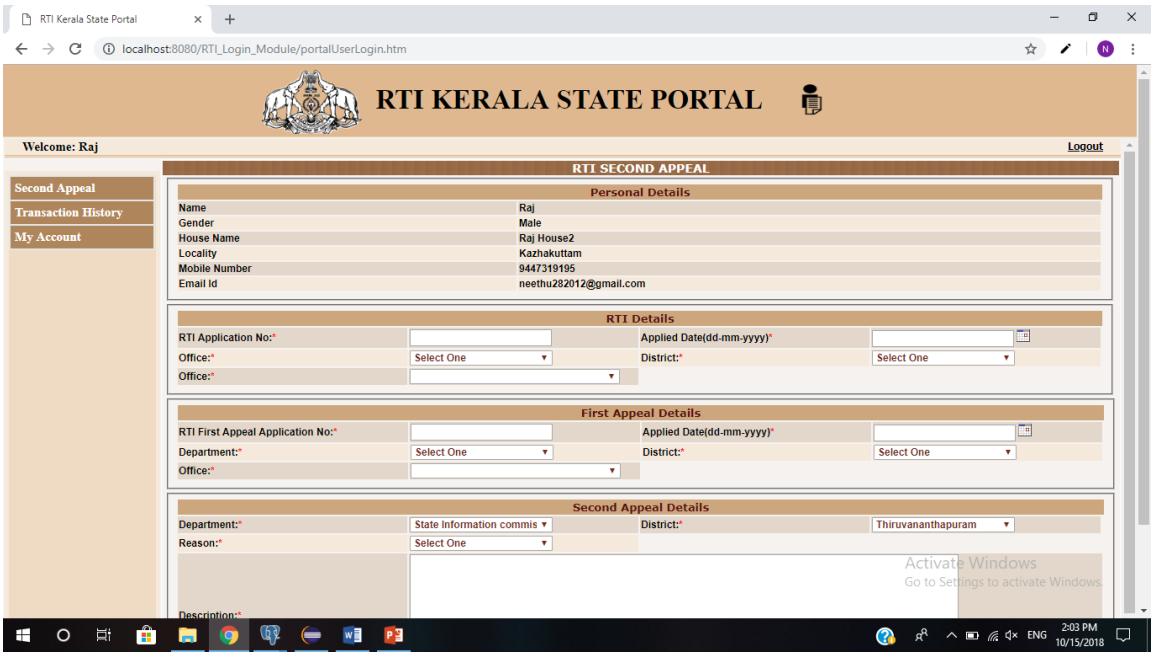


Fig. 8.2: Portal user apply second Appeal-English

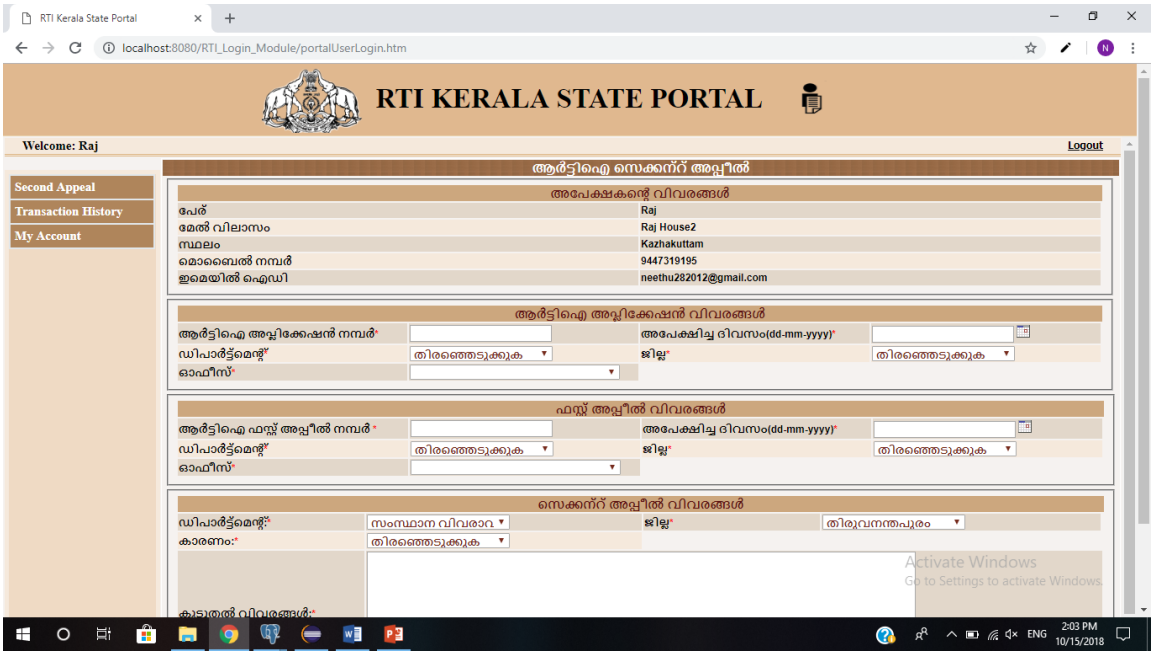


Fig. 8.3: Portal user apply second Appeal-Malayalam

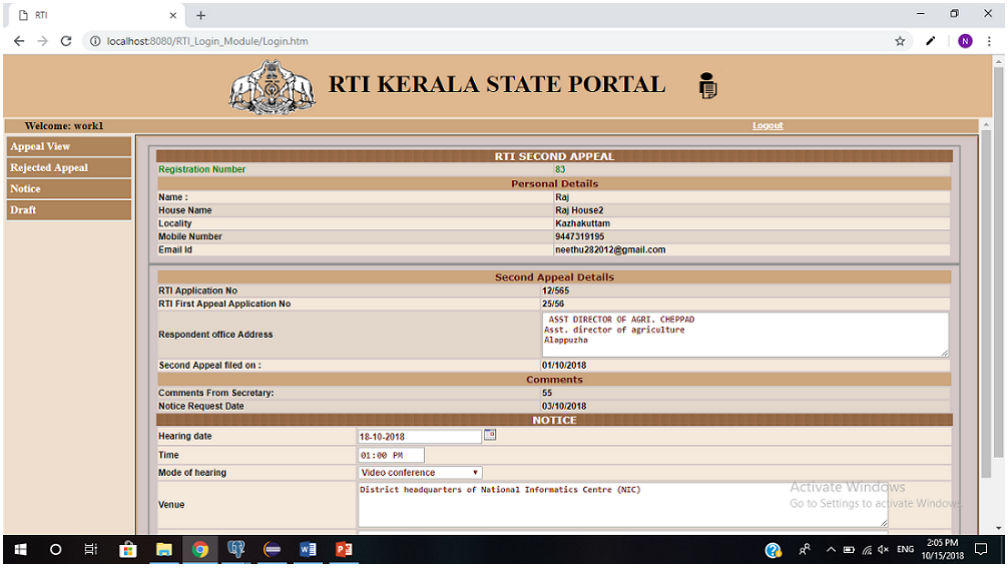


Fig. 8.4: Work desk Notice creation

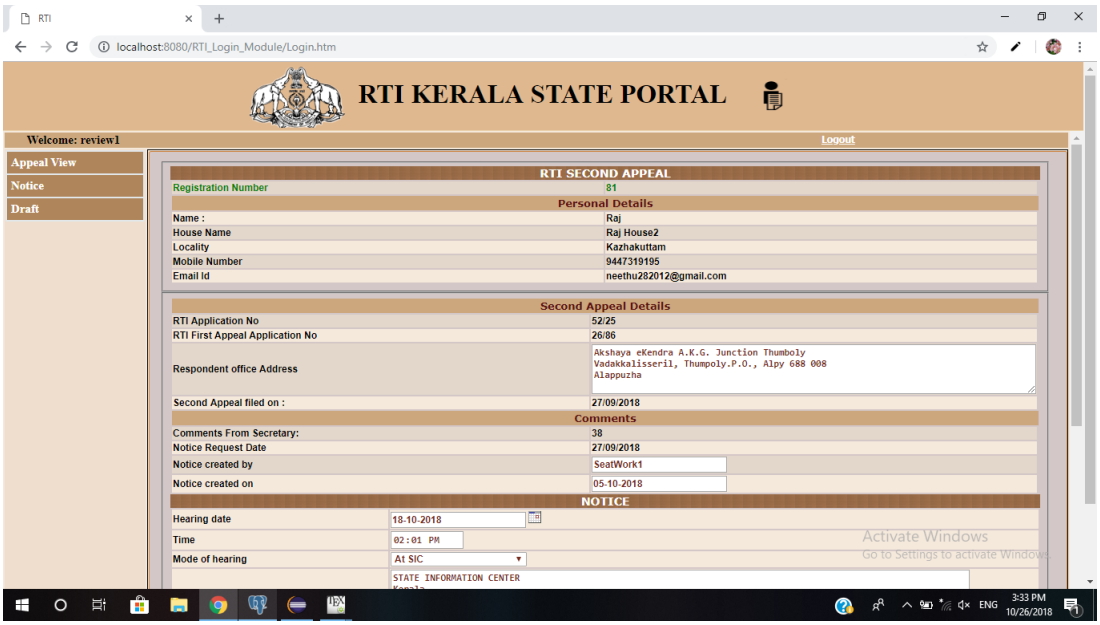


Fig. 8.5: Review Notice verification

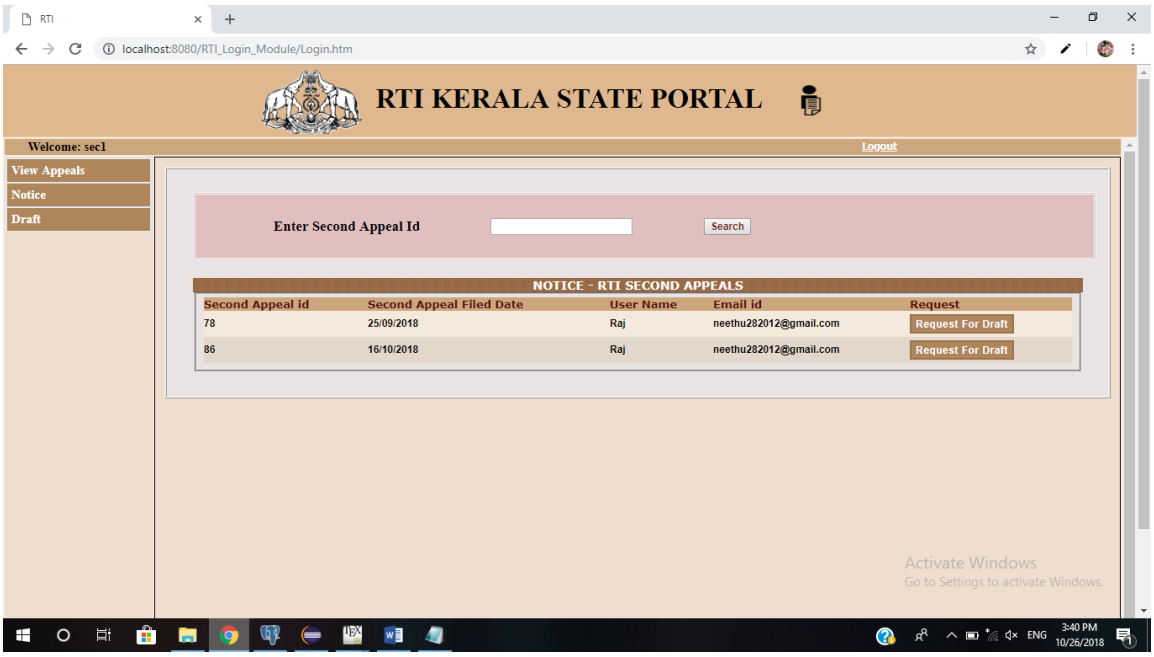


Fig. 8.6: Secretary requesting page for draft creation

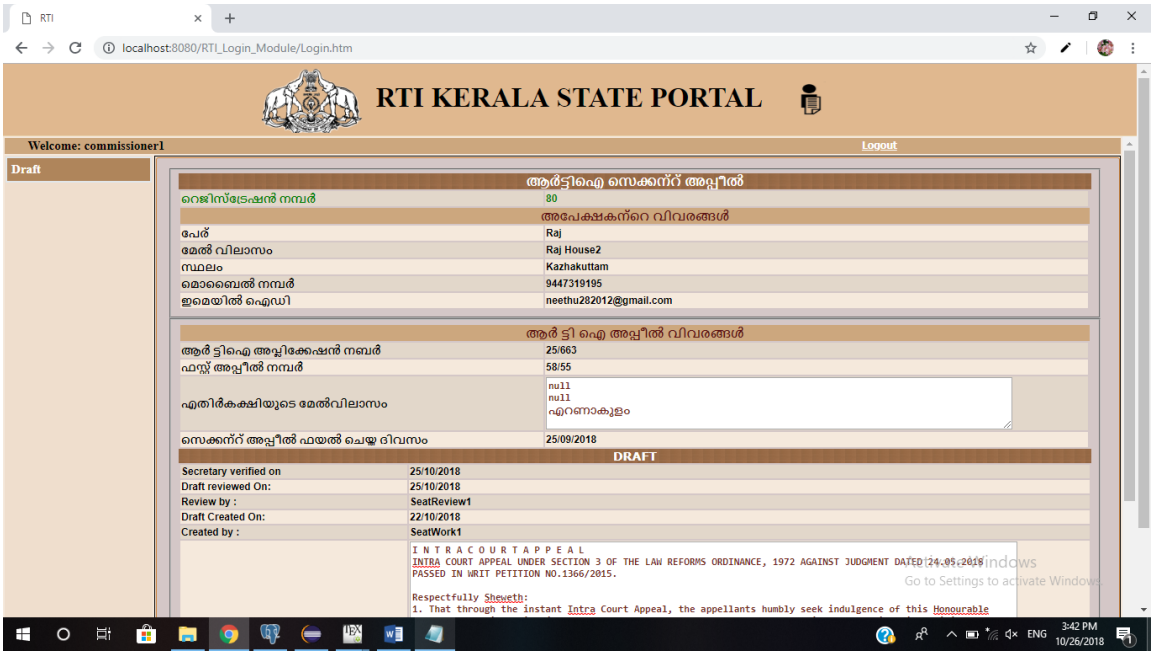


Fig. 8.7: Commissioner approving draft and appeal

BIBLIOGRAPHY